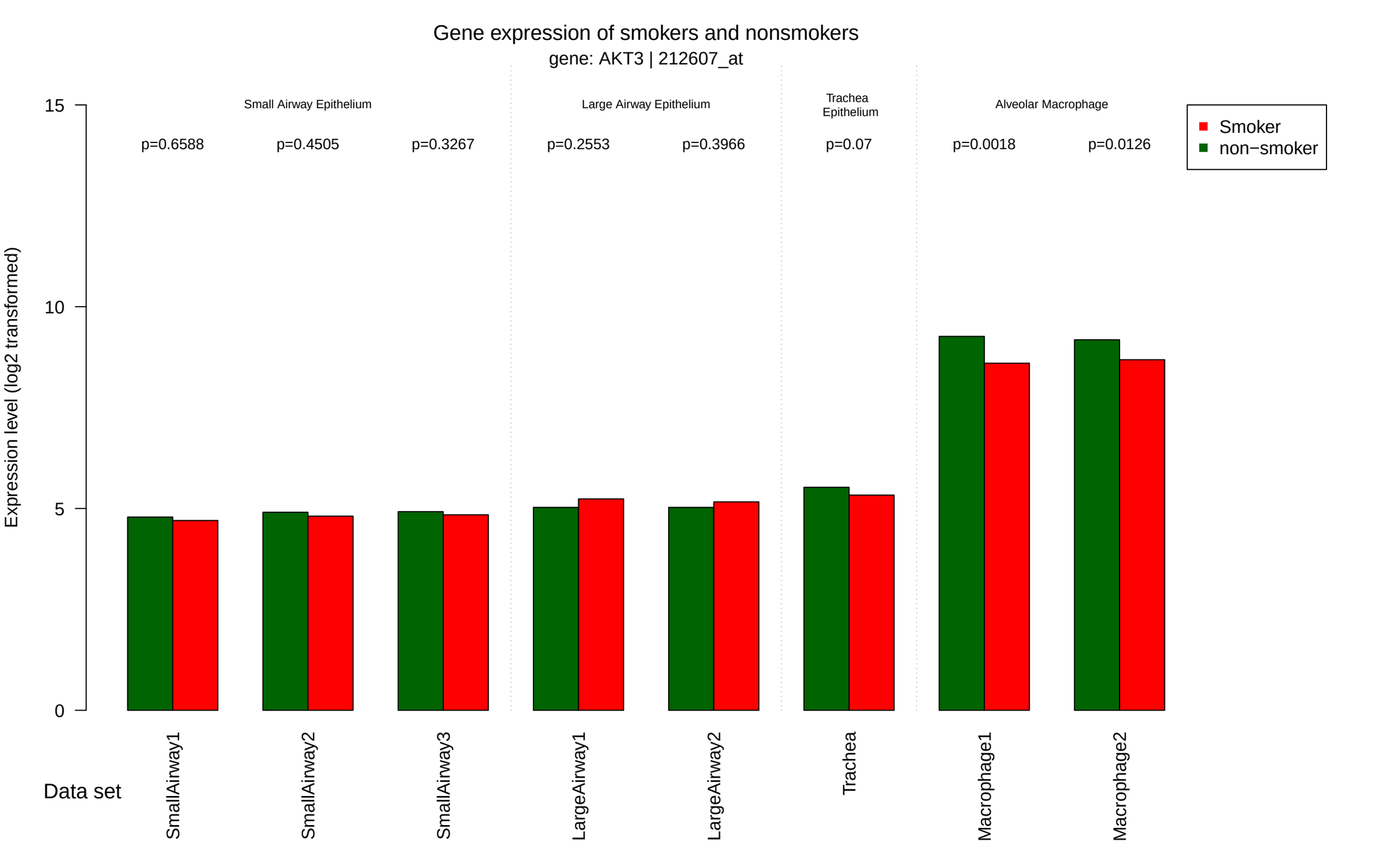
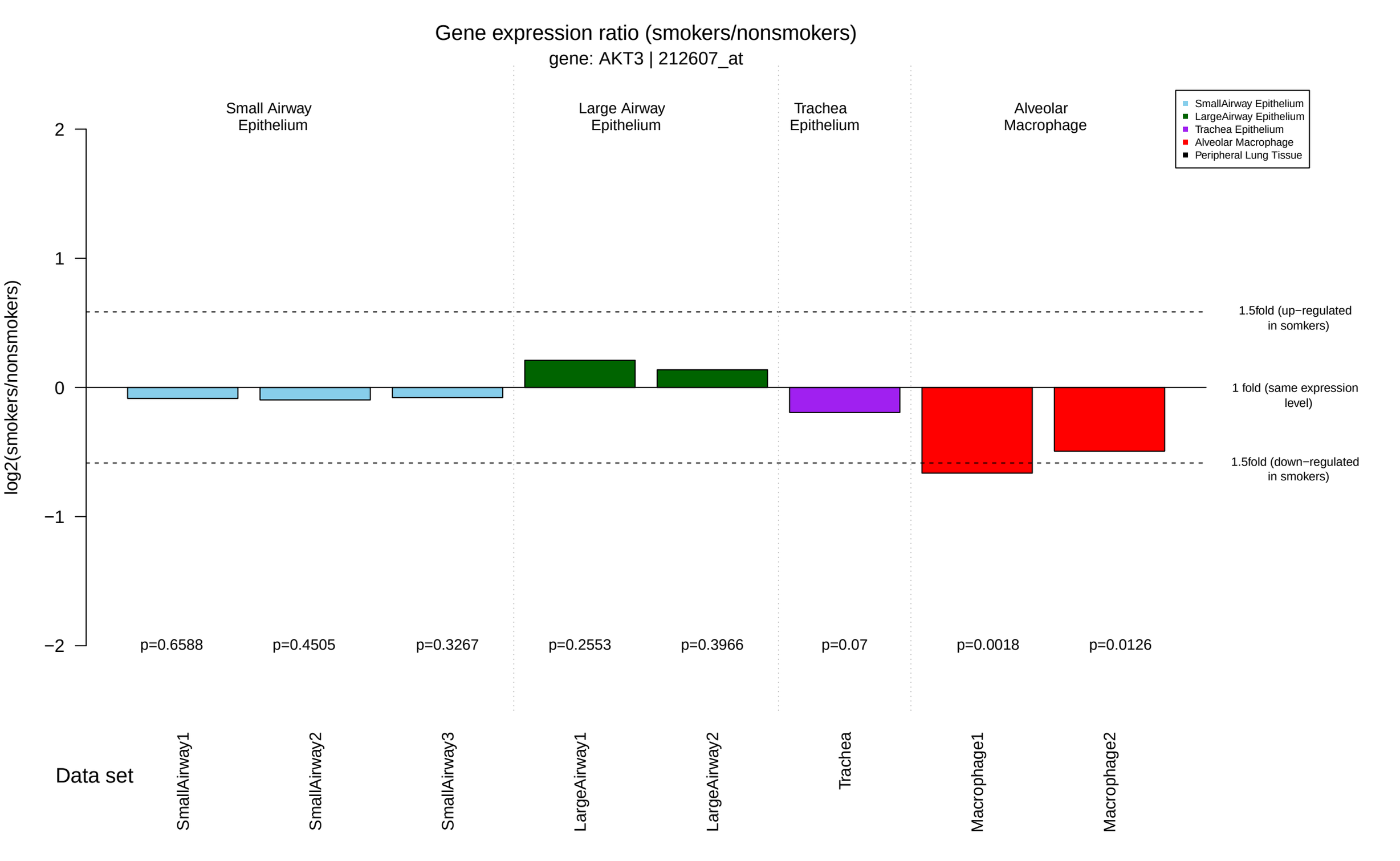
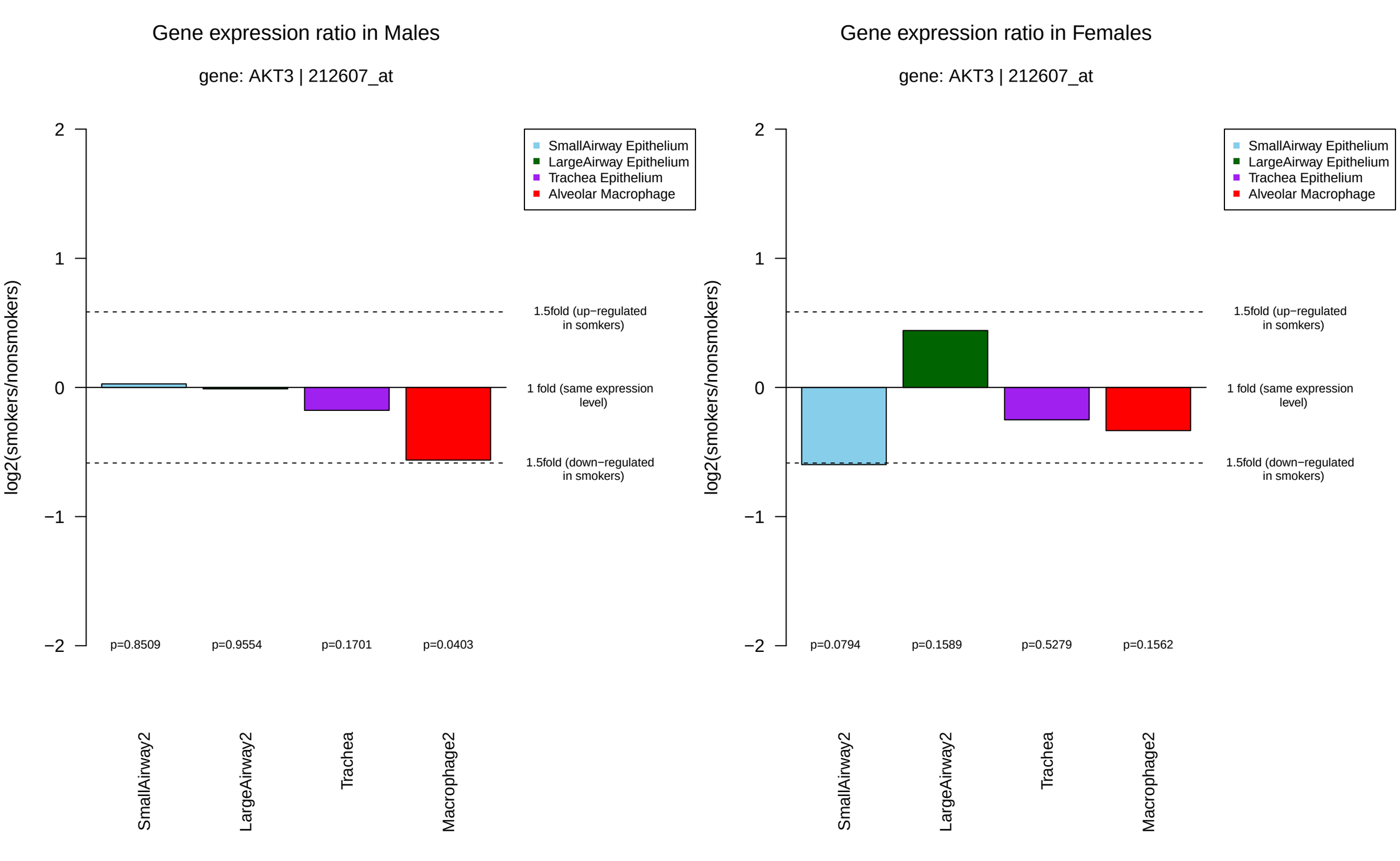
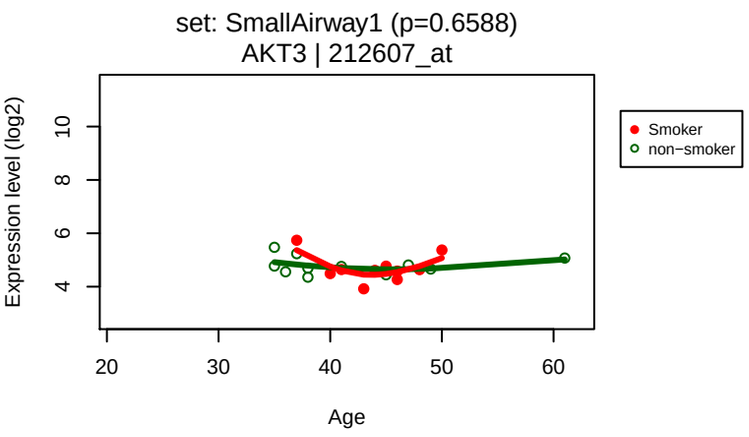
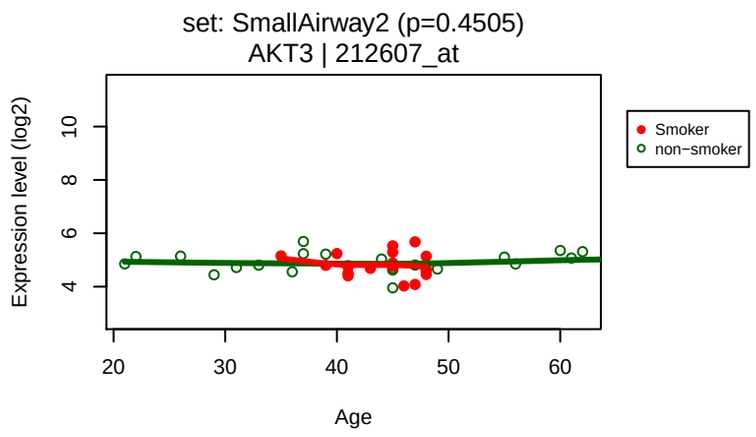
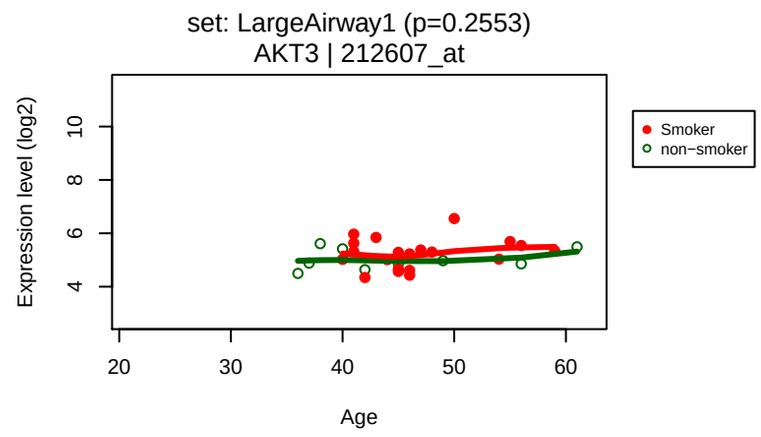

Gene expression of smokers and nonsmokers
gene: AKT3 | 212607_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6588
p=0.4505
p=0.3267
p=0.2553
p=0.3966
p=0.07
p=0.0018
p=0.0126
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

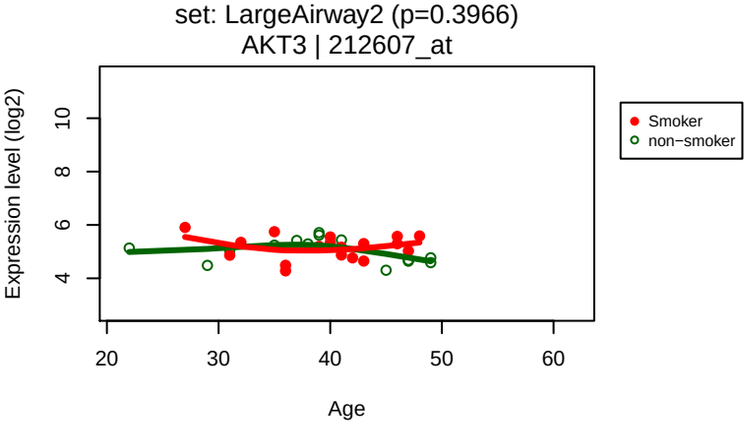

Gene expression ratio (smokers/nonsmokers)
gene: AKT3 | 212607_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6588
p=0.4505
p=0.3267
p=0.2553
p=0.3966
p=0.07
p=0.0018
p=0.0126
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

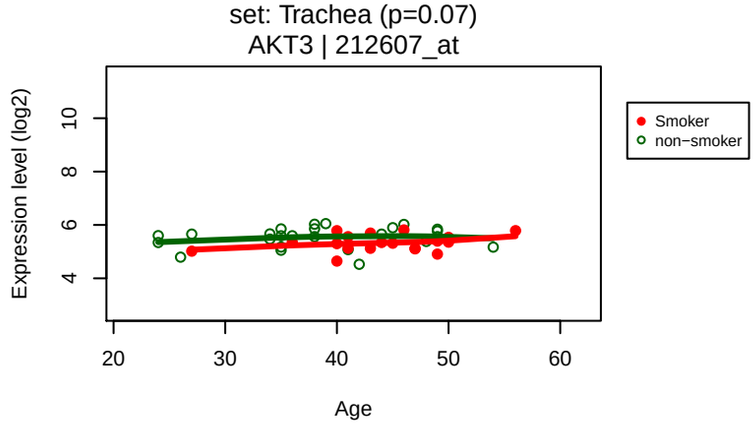

Gene expression ratio in Males
gene: AKT3 | 212607_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8509
p=0.9554
p=0.1701
p=0.0403
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKT3 | 212607_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0794
p=0.1589
p=0.5279
p=0.1562
SmallAirway2
LargeAirway2
Trachea
Macrophage2

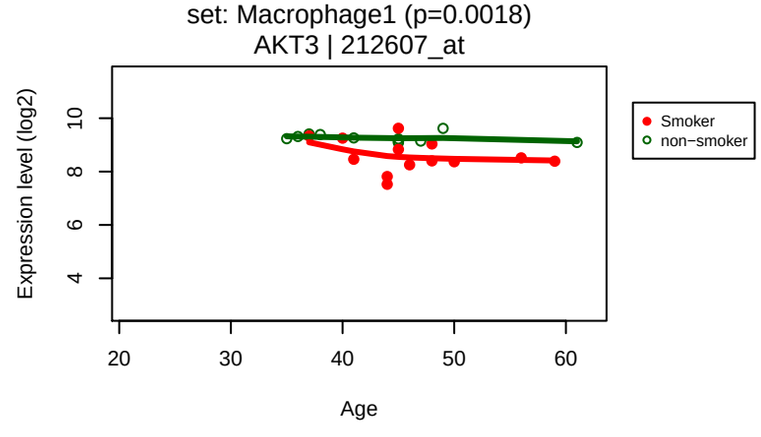

set: SmallAirway1 (p=0.6588)
AKT3 | 212607_at
Expression level (log2)
Age
Smoker
non−smoker

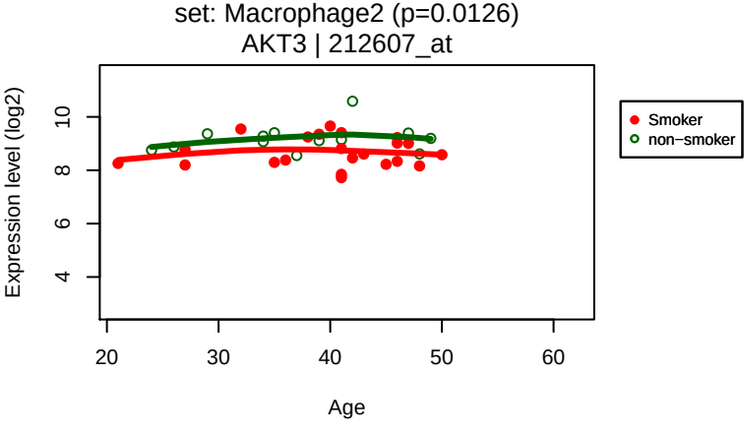

set: SmallAirway2 (p=0.4505)
AKT3 | 212607_at
Expression level (log2)
Age
Smoker
non−smoker

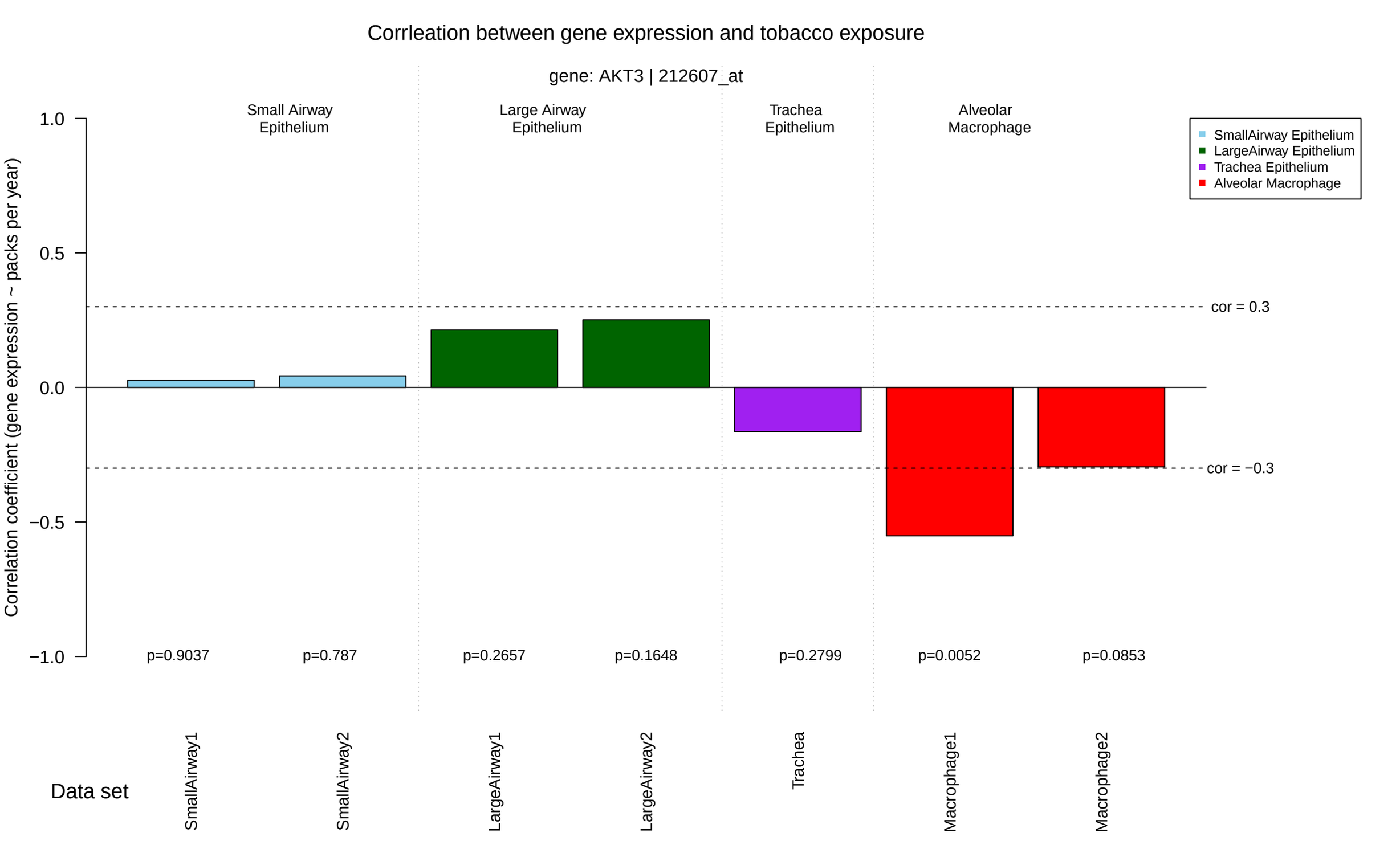

set: LargeAirway1 (p=0.2553)
AKT3 | 212607_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3966)
AKT3 | 212607_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.07)
AKT3 | 212607_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0018)
AKT3 | 212607_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0126)
AKT3 | 212607_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKT3 | 212607_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9037
p=0.787
p=0.2657
p=0.1648
p=0.2799
p=0.0052
p=0.0853
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2